RahScan — AI-Powered Pothole Detection & Road Damage Reporting System

Final Year Project Proposal

(BSCS)

By

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ABSTRACT

One of the key issues of public safety in Pakistan is the breakdown of the road infrastructure which leads to thousands of accidents each year. Big cities such as Lahore, Karachi, and Faisalabad are extremely vulnerable to potholes and damages on pavements. There is no centralized or data-driven system in place to monitor the detection of potholes, their reporting, or planning to be maintained. The existing techniques use manual surveys, which are not quick, expensive, and extensive. There is no organized system that the citizens have to report road damage. Authorities are reactive rather than being predictive and evidence-based in their planning.

The pothole detection and reporting system suggested by RahScan is a full-stack web application powered by AI. It brings citizens, AI processing engine, and administrative authorities together in a platform. The images uploaded are processed in real time by a YOLOv8 model trained on RDD2022 and local data. The system identifies potholes, boxes them, categorizes severity and provides confidence scores. These are all saved in PostgreSQL and visualized in dashboards that contain heatmaps and analytics. A prediction of pothole breakdown module is a lifetime prediction that uses severity, traffic and weather data to determine the likelihood of a pothole deteriorating. A predictive maintenance module detects road failures that happen frequently and predicts repairs and budgets. RahScan combines AI detection, prediction and forecasting into a solution on a larger scale to the infrastructure problems facing Pakistan.

INTRODUCTION

In Pakistan, road infrastructure is a crucial requirement to mobility, economic activity and social security, and it degrades at a very high rate due to heavy traffic, monsoons, low quality of construction and failure to maintain it on a timely basis. Big cities of Pakistan like Lahore, Karachi and Faisalabad are known to have high potholes that cause road accidents, injuries, damage to vehicles, and avoidable deaths. The issue is increased by ineffective systems that citizens do not have a well-developed reporting system, and law enforcement agencies use a slow and manual inspection system, which leads to a reactive maintenance system in which repairs are made only after accidents or complaints have been reported.

The development of AI and computer vision can be regarded as a effective solution, and the models such as YOLO provide an opportunity to detect potholes in real time and predictive algorithms that allow forecasting further degradation. Nevertheless, the current systems are in separation and not adjusted to

the road conditions of Pakistan and they are not integrated with detection, reporting, prediction and decision support systems on a single platform.

RahScan fill this gap by designing an AI-based, end-to-end system that allows citizen reporting, processes the images with the help of a YOLOv8-model trained on the local data, labels the severity of potholes, and displays the data in an interactive dashboard. It also has a Pothole Lifetime Prediction module based on severity, traffic and weather data and a Predictive Maintenance module to predict repairs and budget. RahScan supports a shift towards a data-driven, proactive and reactive road maintenance in Pakistan by integrating detection, forecasting, and reporting.

PROBLEM STATEMENT

One of the urban issues facing Pakistan is degradation of road infrastructure, the increased traffic due to motorcycles, rickshaws and transport vehicles and the monsoon rains is causing roads breakdown at a rate that is exceeding the capacity of municipal authorities to repair the roads. Most dangerous of these consequences are potholes which are responsible to 30-40 percent of the road accidents in the city, in high traffic cities such as Lahore Karachi and Faisalabad only one pothole can result in major accidents, financial losses, and serious traffic jams. But the main problem lies in fact that there exist no systematic, data-oriented way of detecting, reporting, and managing road damage. No central pothole database is in place so when damage occurs, it is only dealt with after accidents or complaints have been received, and usually after a lengthy delay. The citizens do not have a systematic system of reporting and they lose access to real-time information and the authorities depend on reactive maintenance, which is more time-consuming, risky and expensive than prevention methods. The analysis of roads is still manual, which reduces the scope of coverage and does not cover every area, and due to a lack of historical data, it is not possible to detect trends, manage budgets effectively, and provide proactive maintenance, which is an essential gap in efficient road infrastructure maintenance.

All these failures have a far-reaching impact on road infrastructure functionality:

- City Traffic Police Departments - Respond to accidents, but do not have preventive infrastructure monitoring technologies.
- Municipal Authorities (LDA, KDA, FDA) control roads but do not have data-based planning systems.
- Safe Cities Authorities include CCTV networks and no road condition analysis is automated.

- Highway and Communication Departments require organized information to plan and budget.
- The passengers are exposed to accidents, damage of vehicles, as well as delays with no reporting system.
- Research institutions have no good data to analyze and make policies.

These gaps demonstrate the necessity of the AI-based system that allows conducting real-time reporting, detecting, predicting deterioration, and planning proactive road maintenance, which is what RahScan will offer.

LITERATURE REVIEW

The past several years have also seen an increasing number of studies in automated road damage detection because of the universality of deep learning systems and the growing accessibility of real-time object detection models. The main points covered in existing literature are detecting potholes with the help of images under the umbrella of deep learning, reporting damaged roads to citizens, and proactive pavement degradation modeling. Despite the significant contributions of each of the areas, none of the studies has been able to combine all three in one end-to-end operational pipeline especially among developing countries like Pakistan. Asad et al. evidenced that deep learning can provide real-time, AI-on-the-edge pothole detection based on a standard camera image, which does not require dedicated hardware, which poses image-based monitors practically feasible [1]. Subsequent research enhanced the level of detection, with YOLOv8-based models having high precision and recall in real-world settings [3], [4], Rout et al. boosted low-quality image detection using ESRGAN with YOLOv8 [2]. Nevertheless, these works do not go beyond detection and are not related to classification of severity, structured logging, or administrative decision making.

A second area of research is on citizen reporting and crowdsourced monitoring of road damage. Salcedo et al. demonstrated that deep learning with citizen reports could enhance the process of road maintenance prioritization, yet also pointed out that the majority of reporting systems do not categorize the severity of the damage, which undermines repair planning [5]. Similar literature has discovered that most systems that rely on complaints have inconsistent location data, ineffective prioritization based on the volume of complaints, instead of the true extent of damage, and insufficient repair-tracking transparency [6]. Li et al. also suggested crowdsourced method of pothole detection based on image recognition to enhance effectiveness in various road conditions [12]. Although this work reinforces the role of the public, a great deal of all of these systems continues to require additional hardware or an often disjointed procedure which

makes them less practical within the Pakistani context where a basic image-upload system can be extended and utilized at scale.

The third and the relatively less developed field of study is the pavement deterioration prediction and predictive maintenance. Research has indicated that weather factors such as temperature and precipitation and traffic volume are significant predictors of potholes and pavement deterioration [7]. Surveys of machine learning applications in pavements management further prove that the regression models, neural networks, Random Forest, and ensemble techniques can be successfully used in predicting the pavement conditions based on historical maintenance and traffic records [10], [11]. Nevertheless, they are forms of predictive systems that are typically developed in vacuity and are not often linked to actual pothole detection pipelines or administrative displays. Consequently, the use of prediction techniques, though promising in principle, has not been extensively used in direct operations in managing municipal roads.

Generally, there are three gaps that are contradictory to the literature. To begin with, pothole detection systems that are based on deep learning are primarily concerned with recognizing potholes and do not relate pothole detection results to either pothole deterioration or pothole detection maintenance prediction. Second, citizen reporting systems often do not include automated severity analysis which makes them less useful in terms of repair prioritization. Third, predictive maintenance research affirms the significance of traffic, weather and past information, yet these findings do not combine with the real time detection and reporting sensing system. Thus, there is no publicly available piece of work that presents a fully connected platform that integrates pothole detection along with severe classification, citizen-reporting, lifetime prediction, historical pattern analysis, and administrative decision support in a single locally customized system. RahScan seals this divide by integrating these elements into a coherent framework that is tailored to the physical environment of roads and infrastructure of the Pakistani cities [1]–[12].

The critical analysis of the available literature describes that there are three crucial constraints in present studies. First, deep learning-based detection systems, like the state-of-the-art YOLOv8 models, are not concerned with severity-related deterioration prediction or maintenance forecasting as their key issue is to detect potholes. Second, there is no automatic severity analysis in citizen reporting site, which limits it in prioritizing the repairs. Thirdly, although predictive maintenance research argues the significance of weather and traffic information in making predictions with regard to road deterioration, they do not combine these forecasts with real-time detection and administration decision-making devices. Consequently, there is as yet no holistic end-to-end system that integrates pothole detection and severity classification, citizen reporting, lifetime prediction by AI, analysis of patterning and history, as well as an

interactive administrative dashboard, specifically adapted to the specifics of road conditions and infrastructure issues of cities in Pakistan. RahScan competes this important gap by including all these elements into one stable intelligent deployable solution.

PROJECT SCOPE

RahScan undertakes the design, development and operational realization of a complete AI-Powered Pothole Detection and Road Damage Reporting System, specific to mirror the road infrastructure monitoring issues common in Pakistani urban settings. RahScan combines computer vision dimension of detection, a citizen-facing reporting interface, predictive deterioration modeling, historical pattern analysis and an administrative decision-support dashboard into one unified, end-to-end pipeline. The scope will include the technical structure of all system elements, as well as their functional combination to a deployable web application that could serve real users with multiple user roles.

RahScan will:

- Give an easy web-based interface to citizens who can report damage on roads by sending pictures and location through any computer or smartphone.
- Detections are made by processing Live images in real-time with an AI detection model that was trained on large datasets, and fine-tuned on Pakistani road images.
- Locate and classify severity using bounding boxes in the images of potholes, and categorizes them as Small, Medium, and Large.
- Give each detection confidence scores as model accuracy and reliability to administrators.
- Keep all reports, locations, severity levels and timestamps in a structured database to track and analyze.
- Visualize data using an interactive dashboard in a heatmap that illustrates the distribution of potholes across Lahore, Karachi and Faisalabad.
- Allow admins to filter, sort and easily control reports with workflow stages:
  In Progress → Pending → Resolved.
- Anticipate pothole damage by data on severity, traffic and weather conditions to approximate crucial timelines and suggest priorities in repairs.
- To create 30-day maintenance plans and estimate the budgets, analyze the historical data to determine the damage trends and generate the maintenance plans.
- Flag data with incomplete location information or low-confidence detections to indicate data quality and high-risk reporting directions.

RahScan is optimized and calibrated to the Pakistani context of the roads, taking into consideration local traffic frequency, monsoon cycle and the nature of road surfaces, which is typical in Pakistani urban roads network. The variant of the detection model that has been chosen supports real-time inference on ordinary server hardware without the need to have head end infrastructure - thus, RahScan can be practically implemented with the resource limitations presented by Pakistani municipal technology departments. The RahScan package will fully include:

- Citizen-facing image upload, location entry, and report submissions with real-time annotated results of detection sent directly to the user, on submission.
- A deep learning based detection model that learns on a dataset of extensive road damage world-wide, and local Pakistani road imagery, that can detect multi-instances of potholes and their severity in real-time.
- A back processing layer that accepts the image uploads, using the detection model, storing data and providing it to the citizen interface and to the administrative dashboard.
- A relational database that keeps tabular standards of all the reports, detection outputs, location standards, status of historical records, and past road spoilage files.
- The Pothole Lifetime Prediction engine uses visual severity per-pothole deterioration forecasts by combining visual severity scores, traffic level classification and live weather data.
- A Predictive maintenance forecasting system that examines any per-location past damage data to determine the presence of any repetitive failure cycles and produce proactive 30-day maintenance programme with budget projections.
- An administrative interface with an interactive geographic heatmap, filterable report tables, system-wide statistics, and a repair-team assignment as well as a status management interface.

PROJECT DEVELOPMENT METHODOLOGY

RahScan is going to be built upon a hybrid Agile approach, which will be based on the iterative and adaptive principles of Agile and the structured phases of machine learning and AI-based projects. Such a method enables constant improvement of detection models, continuous incorporation of system modules and constant review of the system during each development sprint by a supervisor. Every sprint will involve planning, development, testing and evaluation wherein feedback of previous stages will be methodically introduced with future modules being constructed over, them.

1. Dataset Acquisition and Preparation

The basis of proper detection of potholes in Pakistani road setting is a high-quality, representative dataset. The first training dataset will be the RDD2022 Road Damage Dataset 2022 that has over 47,000 annotated road damage images that have been gathered in more than 10 countries. This will be complemented with 200-300 road images taken in Lahore which will introduce new and Pakistan-specific road textures, potholes shapes and lighting conditions, which are not predominant in the world datasets. The local contribution represents a valuable addition to the research in the field.

- Images will be annotated with the help of the Roboflow, and it will be employed to maintain the version of the dataset, as well as implement several augmentation tools, such as rotation, brightness altering, blur, horizontal flipping, and simulated weather conditions, including rain and haze.
- The augmented dataset will be divided to both training and validation and test data in the normal proportions to guarantee credible model assessment.
- A variety of the data across types of roads, time-of-day, and weather conditions will be given priority to have the maximum generalization on the real world of the trained model.

2. Model Training and Optimization

The main architecture that will be used is the YOLOv8, as the architecture offers a good trade-off between inference speed and detection accuracy which is essential in a real-time processing of images given by citizens. The model will be trained in Google Colab with free and easily available results in the form of GPUs, so the training is repeatable without the need to establish institutional computing resources.

- YOLOv8n and YOLOv8s variants will be trained and those will be evaluated, and then the optimal model at the cost of the mean average Precision, inference latency, and hardware cost choice will be adopted.
- Hyperparameters tuning will be done amongst learning rate, image resolution, batch size and training epochs with results recorded systematically to fit the most appropriate configuration.
- To measure the model performance, standard metrics will be measured such as mAP@0.5, precision, recall, and F1 score, and the cases of failures will be analyzed to make needed improvements to the dataset.
- The weights of the trained models will be then exported as a form to be incorporated into the backend inference pipeline.

3. Severity Classification and Confidence Scoring

After detection, a severity classification layer, which will be based on rules, will be applied on each identified pothole case. An inference is lightweight and interpretable, this layer works with the bounding box outputs of the detection model, without the need to have a separate classification network.

- The area of every bounding box that is detected will be computed as a fraction of the overall image frame.
- Potholes that will be considered to be Small, Medium, and Large and marked as needing immediate attention will occupy less than 5 percent of the image, between 5 and 15 percent, and above 15 percent respectively.
- The output confidence score of the YOLOv8 model of each detection will be stored and kept with the severity classification and will offer administrators a reliability measure of each reported detection.

4. Backend Development and Database Integration

The backend will be used as the central processing and data management component of RahScan by bringing the communication between the citizen interface, the AI detection engine, the database, and the administrative dashboard. It will be built in Python with its FastAPI, since it is highly performant, asynchronous processing is built in, and can easily be linked with machine learning libraries in Python.

- RESTful API endpoints: RESTful API endpoints will be used to receive image submission and AI inferences, to serve reports, to filter report retrieval, to update statuses by an administrative user, to deliver heatmap coordinates, and to serve prediction results.
- All the reports with image metadata, the identified number of potholes, their severity levels, confidence level, geographic point, submission time, and repair status will be stored in a PostgreSQL relational database.
- The database schema will be designed to enable real-time querying to the dashboard and historical aggregation to support the Predictive Maintenance module.

5. Pothole Lifetime Prediction Module

This module will respond to a question, which is, How quickly will a pothole, which is presently being identified worsen to an urgent condition? so that authorities need not only prioritize repairs related to the

existing greater severity of a given pothole, but also consider the projected urgency to put that individual pothole in context. It is applied as a regression model which is trained on a mix of pothole attributes and environmental factors.

- Predictive features will also include classification of current severity, estimated bounding box size, road type classification depending on the current reported location, current density level of traffic, and real-time weather flow data that OpenWeatherMap API retrieves such as extent of rainfall possibility and temperature, and seasonal indicators.
- Scikit-learn will be used to train a Random Forest Regression model which will utilize synthetic training data based upon empirically based assumptions of deterioration in cases where the actual historical record is not available when the project commences.
- There will be a predicted number of days to the pothole taking on critical severity, a recommended repair deadline; and a plain-language explanation of the urgency level, which will be produced by the module.
- All predictions will be provided in the administrative dashboard and the respective report entry.

6. Predictive Maintenance Module

The Lifetime Prediction module predicts the future status of individual identified potholes but the Predictive Maintenance module forecasts the proactive maintenance prediction at road segment level based on past records of damage but before the next damage can be visually observed.

- To predict the recurrence patterns of damage in individual road segments that have collected enough historic report data, facebook Prophet time-series forecasting library will be used to model the forecasts.
- The module will find road segments with repeat damage at certain times and create a 30-day forward maintenance prediction with a segment indicating where a new pothole will occur statistically during the forecast period.
- Output will consist of prioritized maintenance schedule based on urgency - Urgent, Plan Repair, and Monitor, and a budget estimate of the repairs to be made within the predicted period.
- In the initial phases of deployment similar to the scarcity of historical data, the module will be used on artificial historical records created to replicate realistic recurrence patterns.

7. Administrative Dashboard and Heatmap Development

The web-based interface that will be created as the administrative dashboard will be a web application built in React.js: the interface where municipal authorities, as well as road maintenance departments, will communicate with all RahScan outputs.

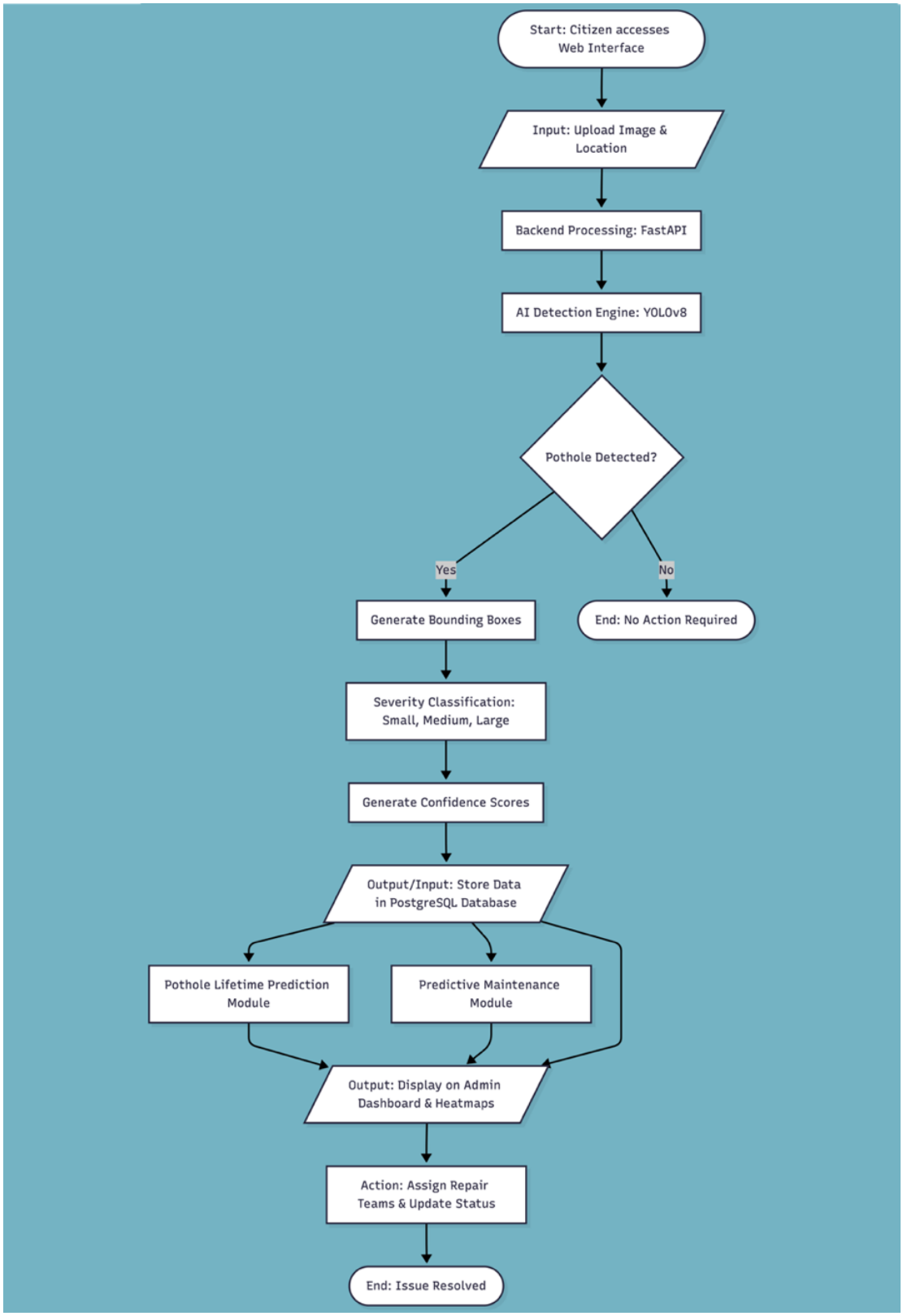
- As an interactive geographic heat map, the map will utilize Leaflet.js and display the density and severity distribution of reported potholes within Lahore, Karachi, and Faisalabad on a map.
- Report management table will have a filter and sort functionality in terms of city, severity level, date range, and repair statuses.
- System-wide statistics panels will present system-wide data such as the number of reports received, reports by severity, and report resolution rates and weekly and monthly trend analysis.
- An interface to repair assignment: The administrators will be able to log repair team assignments and update the report status directly on the dashboard.

8. Testing, Validation, and Continuous Improvement

Intensive testing will be carried out at both module level and the integrated system level to guarantee accuracy, reliability and ability to be applied in the real world before the final delivery.

- All backend API endpoints, severity classification layer, the prediction model inference pipeline, and the database query operations will have unit tests.
- Integration testing will be used to confirm the end-to-end process of citizen image submission to AI detection, storage into the database, prediction generation, and displaying on the dashboard.
- The detection model will be tested on a held-out test set of actual Pakistani road images that were not shown during training and the results will be reported in the form of mAP and precision and recall besides a qualitative analysis of the failure cases.
- At the end of every sprint cycle a reviewing session with supervisors will follow where the results can help to improve model parameters, make the prediction logic and the dashboard easier to use before the next sprint starts.

Flowchart


Start: Citizen accesses Web Interface
Input: Upload Image & Location
Backend Processing: FastAPI
AI Detection Engine: YOLOv8
Pothole Detected?
Yes
No
Generate Bounding Boxes
End: No Action Required
Severity Classification: Small, Medium, Large
Generate Confidence Scores
Output/Input: Store Data in PostgreSQL Database
Pothole Lifetime Prediction Module
Predictive Maintenance Module
Output: Display on Admin Dashboard & Heatmaps
Action: Assign Repair Teams & Update Status
End: Issue Resolved

PROJECT MILESTONES AND DELIVERABLES

The development of RahScan shall be implemented on a systematic six-month program and will be divided into consecutive stages that will build on each other - starting with the preparation of the dataset and the training of the models and proceeding to the integration of the back-end and the creation of the predictive modules and then to the creation of the dashboard and the ultimate system test. Where possible with parallel work streams, the activities will be planned concurrently so as to keep the pace of development without any loss of quality. Each stage will be addressed with a supervisor evaluation prior to the next stage.

Phase 1 (Month 1): will be primarily dedicated to the creation of a high-quality and Pakistan-specific training dataset, which will also entail the download of the RDD2022 and local Karachi and Lahore image collection, annotation, supplementation, and validation of the dataset.

Phase 2 (Month 2): The training of the YOLOv8 model on the pre-ready dataset with the help of Google Colab GPU resources. Both the variants of YOLOv8n and YOLOv8s will be tested, optimization of hyperparameters will be carried out and the final model will be chosen and saved depending on the evaluation metrics.

Phase 3 (Month 3): FastAPI backend development, implementation of all the necessary RESTful endpoints, development of the Postgres database schema, and addition of the trained detection model into the inference pipeline in the backend and the layers of severity classification and confidence score.

Month 4 (Phase 4): entails the development and integration of the Pothole Lifetime Prediction engine, as well as the Predictive Maintenance forecasting module into the backend, integrating the weather API, training the Random Forest regression model and setting up Facebook Prophet as a time-series prediction model.

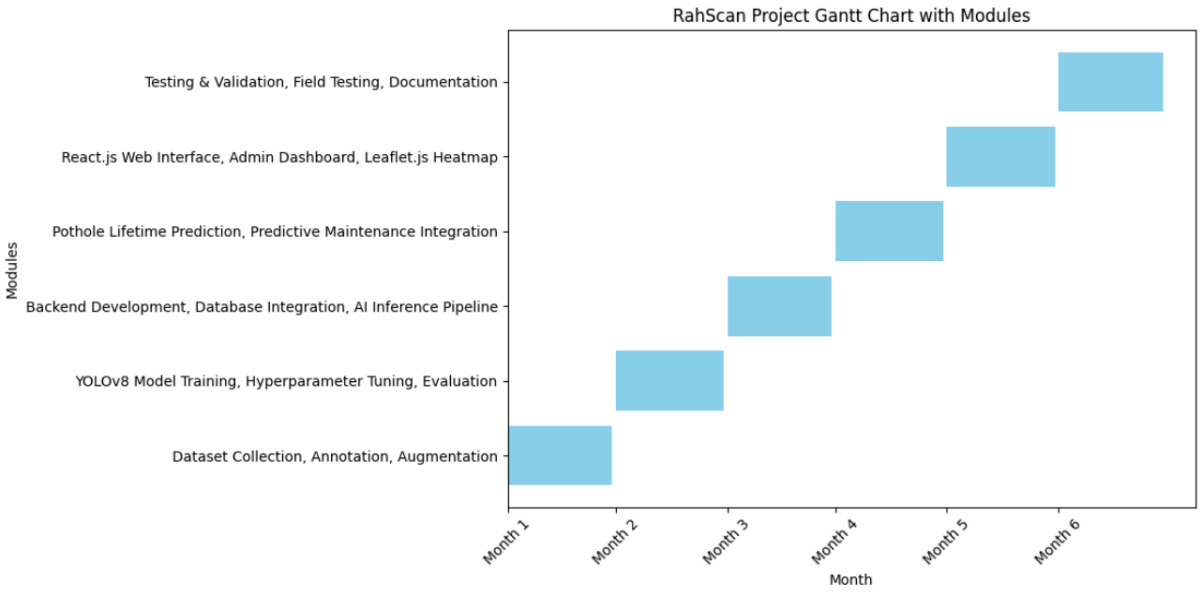
Phase 5 (Month 5): includes the citizen facing React.js web interface and the administrative dashboard, with the Leaflet.js geographic heatmap, filterable report tables, real-time statistics panels, and repair status management interface.

Phase 6 (Month 6): involves detailed unit and integration testing of all RAHscan modules, field testing of the detection model using invisible Pakistani road images and final preparation of the project documentation and demonstration.

Major Deliverables

- An annotated and curated training set with RDD2022 images and supplemented with Pakistani road photographs collected locally that can be trained with YOLOv8.
- A trained and optimized YOLOv8 detection model that can detect potholes in real-time, localize the bounding box, classify the severity, and score the confidence.
- A back-end capable of running a full-fledged RESTful endpoint, database-integrated, and full-fledged AI inference pipeline.
- A Pothole Lifetime Prediction engine combining visual severity with traffic density and real-time weather information to predict per-pothole deterioration.
- A Predictive Maintenance module that produces proactive 30-day maintenance forecasts of road segments that are maintained with budget estimates based on historical trends of damage.
- An administrative dashboard with an interactive heatmap, report management tables, which can be filtered and system-wide statistics, as well as a repair assignment interface.
- A user-friendly web interface allowing image upload, location input and real-time annotated detection outputs.

Gantt Chart



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